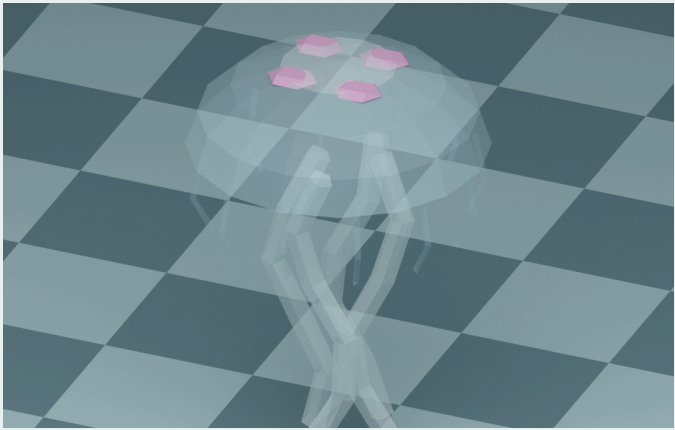


MOON JELLY · THE ART OF THE PULSE

Aurelia aurita · Contraction, recovery and coasting

STUDY checked observation + conditions **SCHEMATIC** explanatory drawing **GAME** chosen animation parameters

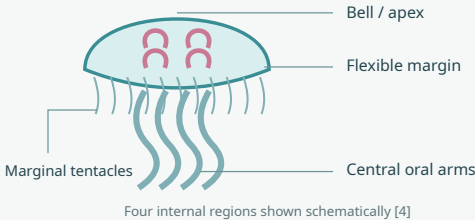


BLENDER · exported model
Bell 22% opacity · margin/arms 32% · fringe 12% · internal motifs 72%.



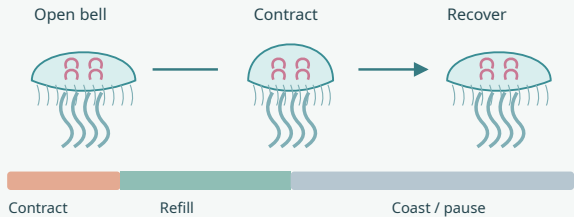
IN GAME · actual OpenGL capture
Six translucent jellies, native bell deformation and trailing appendages. Alpha compositing; no refraction.

A Anatomy



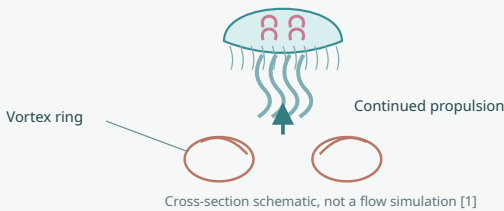
An oblate bell, a flexible edge, short marginal tentacles and longer central oral arms. Distinguish the fringe from the arms. [4]
SCHEMATIC

B Contract → recover → coast



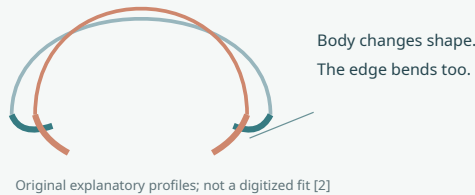
A quick active stroke, recovery of the bell and an interval before the next stroke. Phase widths are adopted game choices. [1, 2]
SCHEMATIC / GAME

C Travel beyond the stroke



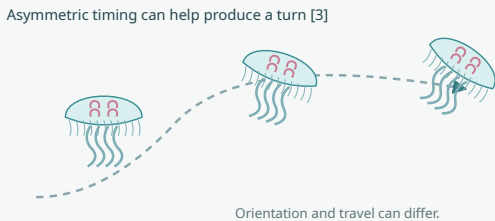
The moving water beneath the bell can continue producing thrust after active bell motion has stopped. [1]
STUDY + SCHEMATIC

D A flexible margin



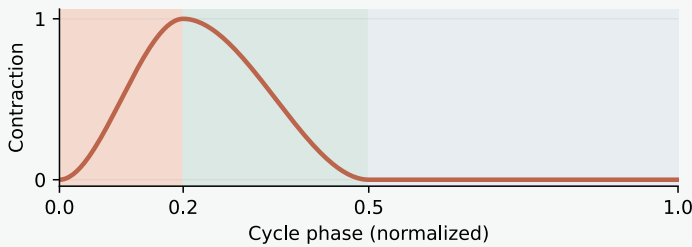
The margin does not simply follow the rest of the bell. Uniform scaling misses part of this deformation. [2]
STUDY + SCHEMATIC

E Turning and trailing



Different timing on opposite sides can accompany turning. Trailing appendage lag is a game choice, without a measured delay here. [3]
STUDY + GAME

F From observation to animation



GAME animation: 20% contract · 30% recover · 50% coast. A 4.0 s player cycle; residents use separate pulse and drift clocks.
GAME · COMPUTED ILLUSTRATION, NOT EXPERIMENTAL DATA

Natural timing varies [4]

STUDY: 0.24 ± 0.11 Hz in endogenous vertical swimming trials (n = 8; resting bell diameters 13–19 cm). This is a study-specific result, not a universal pulse rate.

Keep the simplification visible

GAME: margin-weighted radial contraction ≤18%; weighted height coefficient 28%, with a pinned apex. The compact rig does not reproduce full margin asymmetry or calculate the surrounding fluid flow.

Primary research · original diagrams, no copied experimental figures

[1] Gemmell et al. (2013), PNAS.
[Passive energy recapture · DOI 10.1073/pnas.1306983110](https://doi.org/10.1073/pnas.1306983110)

[3] Costello et al. (2024), Bioinspiration & Biomimetics.
[Turning kinematics · DOI 10.1088/1748-3190/ad1db8](https://doi.org/10.1088/1748-3190/ad1db8)

[2] Villanueva, Vlachos & Priya (2014), PLOS ONE.
[Flexible margins · DOI 10.1371/journal.pone.0098310](https://doi.org/10.1371/journal.pone.0098310)

[4] Xu & Dabiri (2020), Science Advances.
[Anatomy / variable pulses · DOI 10.1126/sciadv.aaz3194](https://doi.org/10.1126/sciadv.aaz3194)